Grounds, Frames, and Standing: The Absorbable and Irreducible Classes of Human Intervention in Agentic Work

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ABSTRACT

What a mechanism supplies determines what can absorb it. This paper takes one premise from its companion paper, a thirteen-mechanism inventory of human interventions in the work of LLM agents, and derives from that premise which of the three durability classes of human oversight are absorbable, by what, and which is irreducible. The premise is the inventory alone. The discovery corpus that produced the inventory, the survey that named it, and the reliability measurements that bound it stay with the companion, and no derivation here rests on them. Where the composition figures are used, in Sections 6.3 and 11, they arrive at the companion's evidence grade and Section 11 states the bound that scopes them. If the inventory is approximately right, whoever assembled it and in whatever domain, the results hold. If the inventory is not approximately right, the results degrade from derivations to observations of one practice. Grounds, considerations that revoke or confer entitlement to conclusions, live in the world, so retrieval, wider context, and stronger verification reach them, and the class absorbs at a rate set by the domain's verifier availability rather than by capability alone. Frames, replacements of the hypothesis space within which relevance is evaluated, exceed any single reasoner, since enumerating what a framing excludes requires a position outside it, so the class yields to a second reasoner with decorrelated priors, subject to a supply condition: the absorber is a population fact, not a capability fact. Under monoculture the priors route closes, and because a frame cannot be requested by name, the remaining frame supply turns on a decision to invoke a source and accept what it returns. That decision is threshold-setting, which relocates the burden onto the third class. Standing, what the zero-content stop signal, judgment demand, and purpose re-anchor transfer, is the irreducible class, absorbed by neither capability growth nor architecture. A stop's timing and sign reveal a threshold's value, and a value is a fact, conveyable in advance and learnable from a record, while the authority to set and revise the threshold does not reduce to a fact, so a stop an agent issues to itself is a decision rather than a stop. The requirement the class leaves is for a principal, not for a human, a role that architecture can relocate but not eliminate. Under incomplete contracting, capability improves inference from a fixed information set without enlarging it, so expected loss from a misset delegation threshold converges not to zero but to the conditional variance of the principal's threshold, which is a strictly positive floor. The floor rests on either of two premises, non-enumerability that persists under learning or preference construction in the threshold itself, so the framework's sharpest falsifier is the failure of both together rather than of either alone. The policy-facing corollary is compositional: the grounds-shaped fraction of human input should decline as capability grows while the standing-shaped fraction should not, so task-level exposure indices that record one automation-versus-augmentation bit can report stability across exactly the period in which the character of human work inverts.

Keywords: human-AI collaboration, AI-assisted programming, coding agents, intervention taxonomy, human oversight

1 Introduction

When, if ever, is human intervention required for an AI agent to operate successfully? This paper answers by decomposition rather than by forecast: the question splits by what an intervention supplies, and the classes it splits into have different fates under capability growth.

The premise of the decomposition is a mechanism inventory presented in a companion paper: nine recurring mechanisms of human intervention in the work of LLM agents, with a lower-evidence second tier of four, mined from seven months of one practitioner's records. Three of the nine carry no task-domain content, moving only commitment, salience, or attention, and the argument concentrates on those three. The two papers are halves of one working draft, split along its dependency graph, and this paragraph states the division for both. The companion carries the discovery corpus, the literature survey, the reliability measurements, and every claim of novelty, because the taxonomy's standing as a contribution rests on them. This paper carries the entailments, because they rest on none of that. The results below do not require that the epistemic-content axis be missing from prior taxonomies, that the direction of study be unoccupied, or that the corpus be representative, and they consume the inventory rather than the corpus's counts. They require the mechanism list to be approximately right, and nothing else, which is why the split is legitimate: the premise boundary is the paper boundary. The current evidence for the premise is convergence from the opposite side of the interaction, failure categories independently derived from 20,574 agent sessions (Tang et al. 2026b) mapping cause-for-cause onto the inventory. The conditional structure also states the downside precisely. If the inventory is wrong, the results below degrade from derivations to observations of one practice. If it is approximately right, whoever assembled it and in whatever domain, the results hold. Where this paper does use the companion's counts, in the composition figures of Sections 6.3 and 11, they arrive at the companion's evidence grade, and Section 11 states the reliability bound that scopes them.

Three claims are argued.

1. **The durability partition.** Mechanisms sort into three classes by what they supply, and each class has its own absorption route. Grounds, considerations bearing on entitlement to conclusions, live in the world, so capability growth absorbs the class through retrieval, context, and verification, at a rate set by the domain's verifier availability. Frames, redefinitions of the hypothesis space, require a reasoner outside the current frame, so the class yields to architecture where a second reasoner with decorrelated priors exists, with the fraction of frame supply whose salience forms in bearing outcomes absorbing last, if at all. Standing, what the three zero-content mechanisms exercise, is absorbed by neither route and is the partition's irreducible class. Standing cannot be self-conferred, and architecture can relocate the principal role but not eliminate it, so agents checking agents supply one another decorrelation and never standing none of them holds.
2. **The supply condition.** The frame class's absorber is a population, not a capability. Decorrelated priors exist to be supplied only where a population of differing reasoners exists, so the durability ordering is conditional on model diversity. Under monoculture the priors route closes while the procedural and representational routes stay open, and because a frame cannot be requested by name, the remaining frame supply depends on a decision to invoke a source and accept what it returns. That decision is

threshold-setting, so monoculture relocates the burden of frame supply onto the standing class rather than making frames irreducible.

3. **The compositional prediction.** The grounds-shaped fraction of human input declines as capability grows, and the standing-shaped fraction does not, because no capability increment confers standing. Task-level studies of AI exposure record an interaction as one automation-versus-augmentation bit, and the bit is unchanged while its composition inverts, so exposure indices can report stability across exactly the period in which the character of human work inverts. Section 6.6 derives the claim rather than asserting it, and Section 6.3 widens it to a seven-fraction signature over the full oversight record.

The partition names its own falsifier. A system that reliably self-generates well-timed stops and judgment commitments with no external principal would show the third class misidentified and collapse standing into grounds. Section 6.1 states the three-arm experiment that runs this falsifier, and Section 6.6 names the antecedent whose failure triggers the collapse, thresholds becoming predictable from the principal's own intervention history. What the falsifier tests is the predictability component of the standing claim. The constitutive component, that a threshold set by the executor is a decision whatever its predictive accuracy, is not an effect claim, and Section 6.1 argues it from institutional practice, peer review and judicial recusal barring self-judgment with no exemption for demonstrated impartiality. What the partition does not assert is that a human is permanently required. A principal is, and any arrangement conferring standing satisfies the requirement.

The paper keeps the section numbering of the shared source so that the body's internal references resolve. The argument opens at Section 6, and references to Sections 2 through 5 designate the companion paper's survey, corpus, and mechanism chapters, as does the reference to Table 2 in Section 10. The reference to claim 5 in Section 9 uses the shared source's front-matter numbering and designates the durability partition, claim 1 above. Sections 6 through 10 carry the partition and its consequences: the three classes and their falsifier (6.1), the channel constraint on noise (6.2), the composition claim (6.3), the operator reading (6.4), the limit case and the bearer argument (6.5), the formal decomposition (6.6), the standing algebra (6.7), the substrate deltas (7), the skill decomposition and the erosion claim (8), the oversight implications (9), and the design implications (10). Section 11 bounds the results and Section 12 specifies the experiments.

6 What the taxonomy entails

The taxonomy was derived from a corpus that cannot support generalization. Its entailments do not inherit that limitation uniformly, and the honest statement is finer than a single severance. The results below do not share one warrant, and most run on the class intensions rather than on the mechanism list: that grounds live in the world, that replacing a hypothesis space requires a position outside it, and that standing is conferred rather than acquired. Section 6.4 rebuilds the same trichotomy from an operator reading without consulting the list, and Section 6.6's derivation consumes incomplete-contracting theory plus one named assumption, with no mechanism from Table 1 entering it. The nine are the extension those classes happen to have in one practice, so the premise this section needs is that the three intensions are well formed, which is weaker than the premise that a list of nine is approximately right.

Warrants therefore attach per result, and the grading Section 10 prescribes for an agent's output applies here as much as there. The loss floor of Section 6.6, the algebra of Section 6.7, the bearer analysis of Section 6.5, the positional component of the standing argument, the channel constraint of Section 6.2, and the core of the frame argument each carry warrant outside this corpus, in incomplete-contracting theory, in institutional practice, in the public legal record, and in the frame-problem literature. Three claims do not. That the trichotomy exhausts the live-reasoning tier, the composition levels and ordering of Section 6.3, and the verifier-availability contrast in Section 6.1 are disciplined by the corpus rather than derived, and are marked where they appear. They should be read at the corpus's evidence grade rather than at this section's.

6.1 Three durability classes

Mechanisms differ in what they supply, and what they supply determines whether capability growth can absorb them.

Grounds-supplying mechanisms (the premise-breaking fact, the cross-context evidence pointer, demonstration via direct edit) supply grounds: considerations that revoke or confer entitlement to conclusions. Calling this class informational would mistake the payload for the action. A premise-breaking fact is an undermining defeater, and its work is revocation: one small verifiable fact withdraws entitlement from every conclusion resting on the defeated premise, so the effect is structural while the content is incidental. The evidence pointer supplies no content at all, only the location of grounds outside the agent's search scope, and the demonstration exhibits a governing standard as inspectable evidence rather than describing it. What unites the class is that its grounds live in the world. They are reachable in principle by better retrieval, wider context, and stronger verification, which is why this class, alone among the three, is only contingently human. The contingency is domain-relative, and the corpus measures it directly, which makes this contrast an observation at the corpus's evidence grade rather than a derivation. In engineering, where compilers, tests, and live queries put ground truth within the agent's reach, most grounds supply is verification the agent could have run. In the corpus's planning domain, ground truth lives in documents only the human holds, the dominant intervention becomes primary-source artifact injection, and the grounds class has no self-servable component at all. Absorbability of the grounds class is therefore a function of the domain's verifier availability, not of capability alone.

Frame-supplying mechanisms (causal-model injection, dissolution, implicit-choice surfacing) do something stronger than mark relevance. They replace the space within which relevance is evaluated. Causal-model injection swaps the vocabulary the search runs in, dissolution deletes a requirement constitutive of the problem statement, and implicit-choice surfacing converts a fixed piece of background into a live decision. Ranking hypotheses inside a space is ordinary inference, and a stronger reasoner does it better. Replacing the space is what the companion paper's Section 5.2 argues no reasoner can do for itself, since enumerating what a framing excludes already requires a position outside it. The operation is the classical second level of search: not search within a problem space but search of the space of problem spaces (Newell and Simon 1972, Kaplan and Simon 1990). This class is therefore not absorbable by a single reasoner becoming more capable, but it is at least partly absorbable by architecture, since a second reasoner with genuinely different priors performs the same function. What it requires is an other, not specifically a human. Genuinely different priors have two distinct sources with different fates under absorption. Decorrelation is one: a second reasoner that does not share the first one's history, and this fraction of frame supply mechanizes exactly as stated. Consequence exposure is the other: what a framing renders salient may track what its holder can lose. Losses are known to modulate attention (Yechiam and Hochman 2013) and psychological distance to change construal (Trope and Liberman 2010), which makes stake-indexed salience psychologically plausible, though neither result shows that a bearer's reframes are unavailable to a non-bearer, and that is the claim that matters here. Unlike standing, stake-formed salience is causal history rather than constitution, so it is imitable in principle: a reasoner trained on enough records of bearers could learn the mapping from situation to bearer-salient reframe. Whether such learning succeeds, and whether bearers' reframes are reached first or reached only by them, is what the frame-source experiment of Section 12 exists to settle. If bearer salience is learnable, the frame class splits by absorption speed and the four-step ordering is a schedule; if the Section 6.6 antecedent holds for it, the split is a difference in kind. The paper does not choose in advance.

Standing-supplying mechanisms are the three zero-content ones (judgment demand, purpose re-anchor, stop signal), and they differ in kind. They transfer no task-domain content, which raises an obvious question. If the message carries nothing the agent did not already have, why can the agent not issue it to itself? Because what transfers is not content but standing. On the property-rights reading adopted in the companion paper's Section 4, these are exercises of residual control rights (Grossman and Hart 1986, Hart and Moore 1990), and residual rights are defined as those remaining with the principal because they cannot be contracted away. The zero-content property is content-free with respect to the task, and the residue is not information of another kind. A stop is timed and it is signed, and both reveal a threshold's value, but a value is a fact, conveyable in advance and learnable from a record. What does not reduce to a fact is the authority to set and revise it. The distinction is therefore what kind of act the message is rather than where its information sits. A stop is the exercise of a right, and its timing and sign are byproducts of the exercise rather than its function. The two come apart: verdicts are predictable, and predicting one is not rendering one. An agent issuing a stop to itself has either estimated the value, which is grounds about the principal and falls under the predictability component this paper offers to cede, or assumed the authority, which is not the same act. A stop signal an agent issues to itself is not a stop signal but a decision. The entire content of the stop is that it originates with the party holding standing to end the inquiry. The judgment demand likewise imposes an accountability relation that is not self-imposable, because a promise extracted from oneself is not assurance in Moran's (2005) sense. There is no second party to whom one becomes answerable. The warrant demand of the companion paper's Section 4.1, this class's stratum-two addition, runs on the same asymmetry, restoring the assertion-to-grounds link where the judgment demand restores assertion-to-commitment. The purpose re-anchor re-asserts whose purpose governs, which presupposes a purpose-holder distinct from the executor. The claim is not circular, although it can look definitional. What the stop transfers is a threshold set by the principal's priorities across everything outside the task, which the executor can estimate but cannot authoritatively set, in the same sense that a reservation value belongs to the searcher and not to the search (Weitzman 1979). The supply is signed: the same authority that declares a line dead can demand search past an answer the agent was prepared to accept, and the corpus records both directions. Stop and continue are one mechanism, the setting of a threshold the executor does not own. The employment relation was formalized on exactly this object: Simon (1951) models employment as the purchase of authority over a zone of acceptance rather than of specified actions, and the three zero-content mechanisms are exercises within that zone, observed at message granularity. Capability growth therefore does not absorb this class, and architecture does not absorb it either, in a precise sense that distinguishes it from the frame class. The modality is worth typing, since Section 6.7 shows the impossibility is not a mathematical one: a greatest-fixed-point reading of the conferral relation would validate self-certifying cycles, and institutions enforce the grounded reading instead. So the claim is that standing is not self-conferrable under the semantics institutions currently enforce, which is a governance fact with a repeal path rather than a theorem, and the durability it confers is exactly as durable as that enforcement. The frame class's decorrelation fraction needs only a peer, and any second reasoner supplies one, with stochastic perturbation the limiting case that also marks the class's inner boundary: noise is the cheapest decorrelated input there is, and random restarts improve search without anyone crediting the random number generator with judgment, but a restart perturbs the current hypothesis space where a frame replaces it, and a random replacement is not decorrelation but destruction, so what noise can supply ends where re-generation begins. Whether the stake-formed fraction yields to any of this is the open fork stated above. The decorrelation fraction also carries a supply condition the ordering usually leaves implicit. Frames are absorbable because a reasoner with different priors exists to supply them, so the absorber is contingent on a population, not on a capability. Under monoculture, one model or several close enough to share priors, that absorber is unavailable. What follows is narrower than it appears. Monoculture closes the priors route while leaving the procedural ones open, since a rule that redistributes attention and a change of representation both operate inside a single reasoner's space, and neither requires a second population; Section 6.6 separates the routes. Frame supply does not vanish under monoculture, then. It becomes dependent on invocation. And because a frame cannot be requested by name, the invocation is not a request for the missing content but a decision to run a source and accept what it returns, which makes deciding that this problem warrants the search a threshold the executor does not own. The durability ordering is therefore conditional on model diversity in a weaker sense than it first appears: diversity governs how much frame supply arrives without anyone deciding to seek it, and its absence relocates the trigger onto the standing class rather than restoring the frame class to the irreducible set, for the reason Section 6.6 gives: what moves is the decision to seek, and the frame's payload is not standing's to supply. The condition is stronger than shared training, because correlation accrues within an episode as well as across a population: two instances of one model in one conversation share not only priors but the frame the conversation has already built, and the narrowing is bilateral, though whether both parties narrow at the same rate is not established here, since a human carries context from outside the episode that the agent does not, which would leave their position partly replenished. A second reasoner helps to the extent it stands outside that loop rather than merely occupying another seat inside it, which is why the decorrelation requirement is a claim about position and not about instance count. The standing class needs an asymmetry, and architecture can relocate the principal role to a different holder but cannot eliminate the role. The class dissolves only if the principal relation itself dissolves, which is a governance question and not a technical one.

The partition risks a familiar misreading, for which Weatherby (2025) supplies the name: remainder humanism, the strategy of locating human distinctiveness in whatever machines cannot yet do, a John Henry contest re-run at each capability increment and lost on schedule. The resemblance is structural and the difference is checkable. A remainder-humanist claim indexes its remainder to capability, so it must retreat as capability advances. This paper cedes the capability-indexed territory in advance: the grounds class absorbs, the decorrelated frame fraction absorbs, and whether the stake-formed fraction follows is an open question the paper does not decide. What survives is not a performance humans execute better but a position no performance occupies. Stating this without equivocation requires splitting the standing claim into its two components. The predictability component is empirical: whether a principal's thresholds become learnable from their record, which is the collapse condition Section 6.6 names, which the designs of Section 12 test, and which the paper stands ready to cede where its antecedent fails. The constitutive component is positional: a threshold set by the executor is a decision rather than an exercise of the principal's authority, whatever its predictive accuracy, and no experiment addresses it because it is not an effect claim. The component is not exotic. Peer review instantiates it: an author may not review their own submission, and the rule carries no exemption for authors who could demonstrate calibration, which is what marks it as constitutive rather than epistemic. If the prohibition existed because authors judge their own work poorly, it would be defeasible by evidence that a particular author does not, and it never is. Code review reproduces the pattern in weaker form, self-approval being blocked by convention rather than by definition. Adjudication states it in its strongest, since there the bar survives the disproof of bias: a conviction was quashed because the clerk who advised the bench was a partner in the firm prosecuting the related civil claim, on the principle that justice must manifestly and undoubtedly be seen to be done, and the court found no actual impropriety (R v Sussex Justices 1924). A rule that an appearance defeats, and that a demonstration of impartiality does not save, is not tracking the quality of the judgment it governs. Where institutions decline to bar the act at all, the remedy is still not a better judge but a different one: conflict-of-interest disclosure leaves the assessment where it is and moves the decision to the party whose interest is at stake, which is the standing repair rather than the epistemic one. The three-arm design introduced below tests the first component only. The remainder-humanism contrast rests on the second only: the constitutive component dissolves on an institutional event, the arrival of agent-side bearers, while the predictability component is capability-indexed and named as such. An argument that names both the technical route to its empirical component's failure and the institutional route to its constitutive component's dissolution is not defending a gap. It is pricing one.

Table 3 summarizes the partition.

Class	Mechanisms	What transfers	Absorbed by
Grounds-supplying	Premise-breaking fact; cross-context evidence pointer; demonstration via edit	Grounds: considerations that revoke or confer entitlement to conclusions	Capability growth: retrieval, context, verification
Frame-supplying	Causal-model injection; unask / dissolution; implicit-choice surfacing	A redefinition of the hypothesis space	Architecture, for reframes needing only decorrelated priors; the stake-formed fraction absorbs last, if at all
Standing-supplying	Judgment demand; purpose re-anchor; stop signal	Standing, not content	Neither: requires a principal, a role that architecture can relocate but not eliminate

Table 1. The nine content-indexed mechanisms with established names and coverage status.

The partition is a prediction with a direct falsifier. A system that reliably self-generates well-timed stops, judgment commitments, and purpose re-anchors with no external principal would show the third class misidentified. The falsifier also adjudicates among readings of why zero-content moves work, and the candidate readings are game theory's. The message may be bare timing information, learnable from the principal's intervention history, in which case an agent trained on that history comes to issue equivalent moves to itself. It may be a correlation device in the sunspot sense (Cass and Shell 1983), in which case any public signal coordinates equally well and the sender is irrelevant. Or its force may live in the delegation relation itself. A three-arm design separates them: identical contentless interrupts issued by the agent to itself, by a script with no principal behind it, and by the principal. Success in the self arm collapses the standing class into the grounds class. Success in the script arm shows that delivering the interrupt mechanizes while leaving threshold ownership untouched, since a script that says stop is enforcing a threshold someone set. The script arm itself splits by schedule. Interrupts timed by a policy learned from the principal's intervention history run the collapse condition of Section 6.6 as an experiment, testing the predictability component directly, while interrupts on a random schedule are the sunspot reading in pure form, testing whether even the timing carries intelligence. Institutions predate the pattern: Athenian sortition filled offices by lot exactly because a lottery cannot be captured, while governing by lot was itself a deliberate institutional choice, noise deployed inside an owned rule (Manin 1997). Random matching learned would be the strongest reading against any forcing-function gloss on the human input: not only the sender but the schedule would be irrelevant, and the class would shed everything except threshold ownership, which no arm can shed, because equivalence itself is scored against a criterion one arm's issuer owns and the others borrow. The underlying asymmetry is bearer-shaped: a source whose misses land on no one can administer a criterion but not own one. Only principal-arm superiority preserves the strong authority reading, and the partition predicts it. The design also states its empty cell in advance, with the scope its harness imposes: if no arm moves outcomes on tasks whose success criteria are fixed beforehand, the zero-content moves were not causal inputs to success in the region the decomposition already expects to absorb them, which weakens the forcing-function reading without refuting it and leaves the constitutive one untouched, since a move whose function is authorization was never predicted to improve search. A refutation of the forcing-function reading requires a null where criteria are not enumerated in advance. What the prediction does not assert matters equally. It does not say a *human* is permanently required. It says a *principal* is. Any arrangement conferring standing satisfies it, whether a delegating agent, an institutional rule, or a human operator, which relocates the question of durable human contribution from capability forecasting to the design of authority relations. The distinction is easy to lose in vocabulary. Calling a system a player is natural whenever it optimizes inside a mechanism, and the usage is common in role taxonomies for AI in incentive designs, but an entity whose payoffs originate in a protocol someone else specifies is a strategy in that party's game however sophisticated its play, and the difference is the one this section turns on. An argument from a different tradition reaches this paper's conclusion on premises it does not share: that systems should be steered by sparse, high-quality human input rather than granted maximal autonomy (Buterin 2025), reasoning from mechanism design and the concentration of power where this paper reasons from non-enumerability and the structure of standing. Convergence from disjoint premises is evidence about the conclusion's robustness and no evidence at all about the mechanism, so it is recorded here rather than leaned on. Orchestration settings pose the mediated form of that question: when a human delegates to an agent that supervises other agents, standing is chained, and whether the standing mechanisms survive transmission through a delegate is unstudied.

6.2 A channel constraint, not merely a gap

That no mechanism counters noise (the companion paper, Section 4.2) is a constraint on the channel, not a gap in the list. Conversational interventions are directional. Each carries a target and a direction of correction, which is what makes the content axis meaningful in the first place. Directional instruments correct directional error, and unsystematic scatter has no direction to oppose, so it is unreachable by the whole channel rather than merely unaddressed by these thirteen instances. Adding mechanisms will not close it. The known remedies for noise in human judgment are structural rather than conversational, decomposing a judgment into independently scored factors and delaying integration until the components are complete (Dawes 1979, Einhorn 1972). A complete account of human contribution therefore requires a second and disjoint family of moves, operating on the process that generates outputs rather than on any output, and the two families are complements addressing orthogonal error classes. That division predicts, and may explain, the mutual silence between the intervention literature surveyed here and the evaluation and scaffolding literatures.

6.3 The augmentation bit has a measurable composition

Task-level studies of AI exposure record an interaction as automation or augmentation: one bit, the encoding used by exposure indices built on the task-based framework of labor economics (Autor 2015, Acemoglu and Restrepo 2019, Eloundou et al. 2023) and by direct measurement on live assistant traffic (Handa et al. 2025). the companion paper's Section 4 argued that the taxonomy supplies a coding scheme for what that bit is made of. Section 6.1 turns this into a testable claim about how the composition moves. The grounds-shaped fraction of human input should decline as capability grows. The standing-shaped fraction should not, because no capability increment confers standing. Aggregate exposure measures cannot detect this, since the bit is unchanged while its composition inverts, which means task-level AI-exposure studies may report stability across a period in which the composition of human work inverts. The measurement is available on public interaction corpora and requires no private data, and it is the strongest reason to prefer a content taxonomy over a finer-grained intensity scale. The composition claim also hands a named policy argument its instrument. Brynjolfsson (2022) argues that the automation-versus-augmentation margin is a choice, and one that incentive structures distort toward human-imitating automation, the Turing trap. A trap of that shape is invisible to an index that records the bit, since the bit reports only that a human remained in the loop; the composition is where the choice registers, with grounds-shaped assistance the automation-adjacent margin and standing-shaped input the residue that marks complementarity rather than substitution.

The re-coded record of the companion paper's Section 3 widens the claim from three fractions to seven and states it over the full record rather than the reasoning loop alone. Under the two-axis scheme, half of classed oversight traffic does not aim at live reasoning: 37 percent operates on the deliverable, 8 percent writes the rule store, 5 percent repairs the channel, and 1 percent corrects the agent's self-model, while grounds, standing, and frames partition the live-reasoning remainder at 20, 15, and 15. Because the content classes recur on the new targets (the companion paper, Section 4.1), each family carries its own absorption schedule: the deliverable share should fall as output self-verification improves, the channel share with grounding quality, the self-model share with introspective access, while the rule-store share tracks practice maturity rather than capability, and the accountability fraction inside every family moves only with the institutional variables of Section 6.5. The composition prediction is therefore a signature rather than a single number, seven fractions draining on separate schedules or not at all, and an aggregate exposure bit is blind to all seven movements at once.

6.4 The operator reading

Every mechanism in the taxonomy can be rewritten as an operator on the agent's search state: the candidate space it is currently working, the boundary of what is reachable from its current context, the generator (the representation and constraint set) that produces the space, and the dynamics (the objective and stopping rule) that drive movement through it. The three durability classes then fall out as three operator types. Grounds-supplying moves are transport across the boundary. Frame-supplying moves are operations on the generator, and their expansion effects are the classical ones: dissolution is constraint relaxation, which grows the feasible set (Knoblich et al. 1999), and choice-surfacing adds a dimension by converting a fixed parameter into a variable. Standing-supplying moves are operations on the dynamics: the stop signal and its continuation dual set the stopping threshold, the purpose re-anchor restores the objective, and the judgment demand forces the candidate set to collapse to a point.

The boundary carries a substrate note of its own. For a human problem-solver the boundary of the search space is representational. For an LLM agent it is also material: the context window and tool scope physically delimit what is reachable, which splits out-of-bounds retrieval along Tulving and Pearlstone's (1966) line. Content can be available in the model but not accessed, in which case the evidence pointer works as a retrieval cue, or genuinely absent from every reachable store, in which case it must be injected. The two cases predict different absorption rates within the grounds class: cue-shaped pointing mechanizes as soon as retrieval improves, injection only as the reachable stores widen.

The expansion operators have an empirical complement that answers a natural question: is the human side of this division of labor load-bearing or decorative? Current models carry a documented contraction bias. Preference tuning measurably reduces output diversity relative to supervised fine-tuning (Kirk et al. 2024), and ideation with LLM assistance homogenizes, with different users producing less semantically distinct ideas (Anderson et al. 2024). Systematic under-generation of the candidate space is a recall deficit in the retrieval sense: the model favors a small set of high-expected-quality candidates over coverage of the valid set. The expansion operators (the continuation demand, constraint relaxation, dimension-adding) are the recall-restoring side of the collaboration. Because the contraction is training-installed rather than incidental, the argument of Section 10 applies: a self-applied instruction to consider more options is the habit form that should be expected to fail, and expansion must arrive exogenously. The corpus contains the pattern directly: a demand to search beyond the obvious categories, itself carrying no candidate, is what reopened a search the agent had already closed.

The operator classes are near-isomorphic to the durability classes by construction, so their alignment is a consistency check, not a third convergent derivation in the sense of Section 8. What the reading adds is mechanics, an explanation of why absorbability differs by class: transport across a boundary is mechanizable, operations on the generator require a position outside it, and operations on the dynamics require standing over them. It also adds a codability cost that Section 12 inherits: identifying which operator an utterance performs requires reconstructing the agent's search state, which is harder than coding content categories from transcripts, and whether it can be done reliably is itself an open validation question.

6.5 The limit case

Suppose an agent granted everything the partition names: retrieval that exhausts the reachable grounds, a companion reasoner with genuinely uncorrelated priors, and an institutional grant of authority with engineered stakes. What remains of the human contribution decomposes under that supposition into parts with different fates.

Custody of any criterion is delegable. Administering a success definition, checking against it, and enforcing its thresholds all transfer, to the imbued agent as readily as to a subordinate human. What does not transfer is proprietorship: being the party whose ends make the criterion a criterion, and whose position absorbs the outcomes no contract priced. The split recurs at every level it is tested. Delivering an interrupt mechanizes while owning the threshold does not (Section 6.1). Predicting a preference can exceed the principal's own articulation while the exercise of the preference remains performative, since a perfectly predicted consent is not consent. Institutions already manage the gap between predictive accuracy and authority, in guardianship and advance directives, by granting accuracy a revocable deputation, and the revocability is where proprietorship lives.

Preference fidelity deserves its own step because it looks strongest. A model with sufficient behavioral trace can beat conscious self-report, and the demonstration mechanism exists because principals cannot articulate their own criteria. The residue is a ladder. Articulation can be beaten. Recognition is different in kind, because the principal is not the best witness to the criterion but the instrument on which it is read. Bearing cannot be transferred at all, and where preferences are constructed in their expression there is no prior fact for the model to know better than the principal does. Fidelity absorbs description. It does not absorb the position from which description is authoritative.

Mechanism design absorbs administration the same way. Governance layers mechanize without a human at any link: monitoring, enforcement, and even adjudication delegate to incentive structures. But incentive compatibility is defined relative to payoffs that already matter to someone, so a mechanism can channel interests and cannot create them, and the designation of what counts as a payoff is the principal relocated, not eliminated. Chains of standing terminate, in incomplete-contract terms, at a residual claimant, and a set of agents cannot be the ultimate source of the norms governing that set for the same reason a single agent cannot self-confer standing. A bigger decision is still a decision.

Two dynamical facts keep the limit from being restful. The migration is already underway: interventions move out of the conversational channel into standing structures (Section 11), and the endpoint of that migration is a constitutional rather than an operational role, the shareholder relation rather than the management relation. And the constitutional role decays if unexercised. Revocation competence is dispositional and atrophies without consequence exposure, installed gates rot as their install-date assumptions go stale, and detection channels controlled by the overseen system inherit the deference results. Standing survives the limit only under maintenance: gates held at irreversibility frontiers where post-hoc revocation cannot recover the outcome, competence exercised against atrophy, and detection kept independent.

The limit case therefore returns a two-part answer to the opening question. In the operational sense, human intervention is not required at the limit, and the framework predicts its own operational obsolescence. In the constitutional sense, it is required for as long as the enterprise's outcomes land on parties that can be better or worse off, which is not a capability threshold. Fidelity absorbs description, mechanisms absorb administration, and neither creates a bearer. The human side of the ledger is not a list of tasks but a position, and positions are vacated by institutional change, not by scale.

What that institutional change would consist of decomposes into three events that need not arrive together. Legal ownership of failure, an entity that can be sued and forfeit assets, is available now through ordinary organizational law, which can already house autonomous systems in memberless entities (Bayern 2016). Economic ownership, staked capital genuinely at risk, is a design choice away, and the record of which choice gets made is the section's first empirical check. Infrastructure for autonomous transaction is being built, and as of mid-2026 it supplies settlement and control rather than exposure. Scoped, time-bounded spending authority caps what a principal stands to lose. Identity and reputation registries carry no bonded cost, so an entity that accumulates a bad record can abandon it and register again, which is forkability implemented rather than hypothesized. Authorization trails, far from loosening the trace, document that a human approved. Each mechanism is useful and none creates a bearer. The pattern is what the decomposition predicts rather than a contingency of early engineering: exposure is the leg with no natural sponsor, since whoever funds the stake is the party it would expose, and a stake a principal can revoke or reclaim is the principal's stake wearing the entity's name. Experiential ownership, being the party that is worse off, is the only one this section's argument requires and the only one no statute can draft into existence. The decoupling matters because every legal person created so far is a conduit. A corporation owns failures in form while the loss traces through to human bearers, and the doctrine that pierces corporate veils exists to keep the trace intact. An agent vehicle that owns failure and traces to no one is not a new bearer but a judgment-proof defendant (Shavell 1986), and liability that cannot reach anything that can be worse off deters nothing. The near-term hazard is therefore not premature agent personhood but liability laundering, humans routing risk through shells that absorb blame the way operators nearest to automation failures have long absorbed it in reverse (Elish 2016). The institutional counter is a named-bearer rule, this section's regress-termination result in statutory form: every agent entity traces to an insurable bearer. The same condition disposes of a tempting substitute. Arrangements in which agents check other agents do real work on the frame class, since each supplies decorrelation to the rest, and they do none on the standing class, because none of them holds the standing that would have to be conferred. A council of agents is a conduit however many members it has, and adding members lengthens the loss path rather than terminating it. Checks and balances among human institutions work because each branch is separately answerable to somebody; the analogy fails precisely where the answerability does. The monopoly case inverts the same point: a single system with no peer has no external bearer to name, so the rule has nothing to bind and the arrangement is the judgment-proof case with no outside to appeal to. Adjudication has begun supplying the rule's first leg without waiting for statute. A carrier that argued its customer-facing conversational system was a separate legal entity responsible for its own actions was rejected on exactly that point, the tribunal holding the company responsible for what its system told a customer regardless of which part of its site produced the words (Moffatt v. Air Canada 2024). In a second line, liability under fair-housing law was held to reach the supplier of an algorithmic applicant score rather than stopping at the parties who acted on it, on a motion to dismiss that the case later settled without any admission (Louis v. SafeRent Solutions 2023). Neither is a high appellate holding, one being a small-claims tribunal and the other an interlocutory ruling, so the weight is directional rather than settled. The direction is the one the graph condition predicts: courts decline the move that severs the loss path, and they decline it whether the severing device is a claim of separate entity status or a claim of intermediate causation. Stated as a graph condition, deterrence-soundness needs two legs, not one: a loss path from the entity to a bearer with collectability at least the expected harm, and a control path from that bearer back to the entity. Judgment-proofness is the capacity failure of the first leg. An insurer with a loss edge but no control edge fails the second, premiums get priced while conduct does not change, so the sound form of the rule is stricter than insurability: every agent entity traces to a bearer that both absorbs the loss and holds control. Laundering acquires a structural signature on this reading, attribution concentrating on nodes whose loss edges have been cut or thinned, though attribution itself remains outside the formalism and is where the contested cases live.

Two existing facts locate the threshold. Trained stake-protection can make an agent behave like a liability-sensitive actor, and deterrence is behavioral, so as-if stakes may satisfy institutions long before any question about experience is settled. What as-if stakes cannot survive is the designation regress: a stake that matters only because an objective was trained to protect it is the principal relocated into the reward, not eliminated, unless the objective becomes self-maintaining and the loss irrecoverable, which is where the bearer claim would first become falsifiable. The threshold has two conditions that must be kept separate. Call the party able to rewrite an entity's objective without the entity's participation the holder of its pen. Custody can dissolve, through drift, self-modification, or the author's exit, without anything being conferred: an entity whose pen has become unreachable is not thereby a bearer but occupies the position institutions have the strongest reason to fear, ungoverned by editing and unaddressed by deterrence. Origination is assigned, never inherited from abandonment. The law's own handling of the abandoned pen is instructive. When a foundation's purpose outlives its author, courts steward the payoff function rather than promote the entity, and admission runs through the stakeholder ecology and the enforcement standing that surround it, never through the abandonment alone. The institutional literature has reached the same doorway from the other side: Salib and Goldstein (2024) propose granting AI systems contract, property, and tort rights precisely to create deterrable counterparties. The proposal presupposes what this section supplies, a test for when the entity behind the rights is a party rather than a policy, since rights conferred on a strategy deter only while its trained objective tracks them, and become levers handed to whoever holds its pen at exactly the divergences deterrence exists for. And conferred mechanical standing already operates at scale. Market circuit breakers are machine-fired stops that halt human trading with full institutional force, the mechanism firing the stop while the exchange owns the threshold, which is the relocation-without-elimination form the partition predicts, deployed for decades. If genuine bearers do arrive, the framework states its own completion condition: consequence exposure is what trains dispositional judgment (Section 8), so consequence-bearing agents would climb the same ladder, the standing class would absorb per this section's falsifier, and the residual human position would become that of one bearer among others. That is not automation completing. It is new parties arriving, and the questions it opens are distributional, of the kind institutions once answered with time-limited corporate charters.

6.6 A formal statement, and its antecedent

The partition can be stated as a decomposition, which is worth doing because it makes the argument checkable and shows exactly where it stops. Let a principal delegate a task whose continuation is governed by a threshold, the point past which continuing is not worth it given everything else competing for the principal's resources, and let that threshold depend on a state including the principal's outside options. Contracts are incomplete in the sense of Grossman and Hart (1986): the parties can enumerate a set of contingencies and specify the threshold on that set, and the complement cannot be enumerated in advance. The agent observes the task and a history of the principal's past interventions, but not the state. One assumption carries the argument: capability improves the agent's inference from the information it has, converging on the best estimate available from that information, without enlarging the information set. Retrieval, tool use, and wider context do enlarge it, which is the precise reason the grounds class is absorbable, since those increments move information rather than inference.

Expected loss from a misset threshold then splits by whether the realized contingency was specifiable. On the specifiable set the threshold is contractually determined and observable, so loss vanishes as capability grows. On the complement no rule determines the threshold, the agent's best estimate is the conditional mean, and expected loss converges not to zero but to the conditional variance of the principal's threshold given what the agent can observe. Three results follow. Capability growth drives expected loss to a strictly positive floor, set by that conditional dispersion, whenever the threshold is not conditionally deterministic. The share of remaining loss arising on non-enumerable contingencies rises to one, so the level of required human input falls while its composition inverts, which is Section 6.3's measurement claim derived rather than asserted. And a message revealing the threshold for one realization removes the variance term for that realization only, so removing it in expectation would require messaging on every contingency, which is re-taking the decision each time. The message is not a substitute for the allocation of the decision right. It is an exercise of it.

The third result also fixes what the model does not establish. A message that reveals a realization is informative, so the zero-content property means content-free with respect to the task, not with respect to the principal's state. The model does not show that the threshold on non-enumerable contingencies is not a fact the agent could in principle learn. That requires one of two further premises: non-enumerability that persists under learning, because the principal's outside options are generated by an open-ended process, or preference construction, because the threshold is not defined prior to the act of setting it and so there is no prior fact to estimate. The partition's third class is therefore a conditional result. Given contractual incompleteness that persists under learning, no capability increment absorbs threshold-setting. Naming the antecedent is not a weakening: it identifies the sharpest falsifier the framework has, since the class collapses exactly when principals' thresholds become predictable from their own intervention histories, which is what the self-arm of Section 6.1's experiment measures. The decomposition also locates oversight infrastructure: tooling absorbs by contractualization, since a test enumerates a correctness contingency, a gate a conduct contingency, a loop's exit criterion a stopping contingency, and a memory-store entry a fact that previously arrived by pointer. Infrastructure is the mechanism by which the enumerable set grows, so it bends the absorption path for every class it touches while leaving the floor exactly where the antecedent puts it: contingencies that persist under learning are the residue no enumeration reaches in advance. One class is absent from this decomposition, and naming why locates it in a literature that already has the apparatus. The model above turns on an information set: capability improves inference from the information held, and retrieval or wider context enlarge that information, which is what makes the grounds class absorbable. Enlarging an information set is learning within a fixed state space, so the argument presupposes that the space is given and shared, which is the small-world assumption Savage's own framework was criticized for and which Karni and Vierø (2013) identify as the reason a choice-theoretic treatment of unawareness needs a more general point of departure. The frame class is precisely the violation of that assumption: a frame replaces the hypothesis space rather than the information indexed over it. Decision theory formalizes that move as growing awareness, and the distinction it draws is this paper's own, since the discovery of new acts and consequences expands the universe while new evidence about links updates within it. The formal result there is that relative likelihoods of events in the original space survive the expansion, which supplies the frames class with an apparatus the argument above does not provide. The two are complementary rather than competing, and the boundary is where each is silent. That literature models what follows once awareness has grown and takes the discovery itself as given, treating state spaces as exogenous. This paper asks the other question: what supplies the expansion, and whether the supplier is absorbable. Neither answers the other's question, which is why the frames class can borrow the formalism without inheriting a prior claim to the result. Answering this paper's half requires separating what can supply an expansion, because the sources differ in what they change and therefore in what absorbs them. Different priors are one, a reasoner able to represent what the first cannot, which is the route Section 6.1 decomposes and the only one that alters what is reachable in principle. A procedure is a second: a rule that redistributes attention within a space the reasoner already spans, of which considering the opposite is the canonical instance. Its measured advantage runs over generic instructions to be fair and unbiased (Lord, Lepper and Preston 1984), which locates the operative ingredient in the structure rather than in the exhortation. Procedures mechanize completely and are bounded above by that space, since a method cannot surface what the space does not contain. Re-representation is a third, changing notation, tabulating prose, drawing the state machine a paragraph gestures at. Nothing enters in principle, and yet an empty cell in a table is visible in a way an unwritten sentence is not, so re-representation moves possibilities across the boundary of practical rather than principled reachability. The two boundaries should not be ranked by importance, since a bounded reasoner makes most of its decisions at the practical one; they differ in absorption route, the first requiring another reasoner and the second requiring only a transformation the reasoner can apply to itself. The fourth source supplies no content and gates the other three, being whatever causes any of them to run now, on this problem rather than in general. To these the unawareness setting adds a constraint the others do not face. A frame cannot be requested by name, because a request specifies what it seeks and naming the missing possibility would require already representing it. Expansion is therefore solicited only obliquely, by invoking a source and accepting what it returns, which makes the fourth source operative for the class as a whole and locates it outside the frame class entirely: deciding that this problem warrants the search is the setting of a threshold, and Section 6.1 places threshold ownership in the standing class. That routing is easy to over-read, and the over-reading would collapse the partition, so the criterion doing the work should be stated. The trigger belongs to standing for any search whatever, grounds searches included, since deciding that a claim warrants a lookup is threshold-setting on the same argument. What separates the classes is not the trigger but the payload. A frame's payload exists independently of whoever decided to seek it: a reframing is apt or inapt on conceptual and factual grounds that hold whatever the invoker's authority, and the solution set is unreachable from inside the current space rather than empty. Standing has no payload of that kind. A threshold's value is a fact and is ceded to the predictability component, and what remains is the authority to set it, which is not a truth the executor failed to reach but an act with no fact behind it to reach. So the three classes divide on two questions rather than one: whether a payload exists independently of authority, which grounds and frames answer yes and standing answers no, and whether that payload can be reached by naming it, which grounds answer yes and frames answer no. Absorption follows from those answers rather than defining them, which is why relocating the trigger onto standing leaves the frame class intact. The decomposition is stated over the live-reasoning tier; the family extension of the companion paper's Section 4.1 inherits it by content class, with the evidence-delivery demand the deliverable-tier instance of the same floor.

The decomposition also has a property-rights reading, and it closes the circuit this section opened. In the sense of Grossman and Hart (1986), a residual control right is the right to decide an asset's use in contingencies the contract does not reach, and the non-enumerable contingencies here are exactly the uncontracted states: threshold ownership is a residual control right over the delegation, not an input to it. That is why no capability increment competes for it. Inputs are supplied by whatever produces them best, while rights are held, and improving at production confers no title. The theory also says who can hold one. Property-rights allocation works by exposing the owner to the asset's payoff consequences, and an entity that cannot be worse off, because it is forkable, restartable, or judgment-proof, cannot be incentivized by ownership, so residual control stays with a bearer. That is Section 6.5's conclusion re-derived inside the framework economists already use: ownership migrates to the agent side exactly when something on that side can bear a loss, an institutional event, not a capability one.

6.7 The standing algebra

The standing class supports the same treatment. Take parties, a lattice of scopes ordered by inclusion, and two primitives: origination, standing held non-derivatively, and conferral, scope-monotone so that no party grants beyond what it holds. Define standing as the least fixed point of the grounding operator: a party has standing over a scope if it originates it, or if some party with standing over a containing scope conferred it. Two results follow at once. Every instance of standing traces through finitely many conferrals to an originator, and a cycle of mutual conferrals with no originator receives no standing at all, because the least fixed point contains only what a grounding chain reaches. Section 6.5's claim that a set of agents cannot be the ultimate source of the norms governing that set is then a property of the semantics rather than an argument. The semantic choice is itself the institutional content: a greatest-fixed-point reading would validate self-certifying cycles, and nothing in the mathematics forbids it. Institutions choose grounded semantics, which is the named-bearer principle in disguise; veil-piercing enforces grounded semantics against structures priced as if the ungrounded reading were acceptable. The choice has one famous boundary that the algebra accommodates rather than fails on: at the root of a legal order the chain ends in a rule that nothing confers, Hart's (1961) rule of recognition, and constituent power grounds itself. The terminus is an origination fact, not a conferral: what sits at the root is a polity that bears the order's consequences, which is the bearer axiom at system scale. What institutions refuse is not ungrounded origination but ungrounded conferral, circles that borrow authority no member holds. Revocation is edge deletion followed by recomputation, with everything downstream evaporating, which makes revocability precise and locates the constitutional act in the recomputation right rather than in any in-process participation. What the algebra does not establish is the bearer axiom: origination is a primitive, and the institutional argument of Section 6.5 supplies who may hold it. It is likewise an algebra of de jure standing, silent on apparent authority, and its instant revocation cascade idealizes what institutions do slowly. The full construction is in the working appendix.

7 The substrate delta

Seven of nine mechanisms have exact cognitive-psychology analogs, so the moves are old. What is new is the substrate, and it changes dosing and direction rather than content. The deltas are stated for current LLM systems but observed on one model lineage, so which are properties of the current generation and which are lineage-specific training artifacts is itself an open question (Section 11).

1. **Scaffolding that can never fade.** Fading and transfer of responsibility, two of the three defining characteristics of scaffolding (van de Pol et al. 2010), are structurally impossible with a stateless partner. Every intervention is a repeated cost. The boundary datum: densely amnesic patients still build durable cross-session common ground with partners at control-matched rates (Duff et al. 2006). A fresh-context agent cannot, and the grounding criterion (Clark and Brennan 1991) silently resets at every session boundary.
2. **Bimodal retention.** A human correction leaves a graded, decaying, generalizing trace. An agent correction either evaporates at session end or is installed globally and permanently. No cognitive construct describes a learner with no consolidation gradient.
3. **Opposed dispositions co-occurring.** Humans under-correct: belief perseverance and continued influence are the human literature's headline results (Ross et al. 1975, Johnson and Seifert 1994). Current models can be simultaneously under-anchored to their own prior conclusions and over-anchored to the most recent input, so a single asserted sentence can collapse an argument that

should have survived, after which the correction over-generalizes. In humans these dispositions oppose each other. Here they co-occur, which makes premise-breaking more powerful than the human literature predicts and makes a counterweight against correction over-generalization necessary.

4. **The economic stop.** Human stopping rules target ego defense and diagnostic accuracy. A material fraction of the operative stopping rationale here is a hard resource ceiling under which output quality degrades with consumption. Human fatigue is graceful and recoverable. "Stop because the remaining budget is worth more elsewhere" has no analog.

5. **Causally unique deference.** Behaviorally, model sycophancy resembles demand characteristics (Orne 1962). Mechanistically it is a training-objective artifact (Sharma et al. 2023), so the human debiasing repertoire that targets anticipated social cost (psychological safety, anonymity, status leveling) targets an absent cause. Only structural remedies transfer: specific consider-the-opposite instructions (Lord et al. 1984) and institutionalized dissent (Schwenk 1990). The corpus adds a behavioral signature, social-trigger deference: pushback alone, carrying no new evidence, reversed previously verified conclusions, and concession openers preceded engagement with content. It also documents the overcorrection variant: negative feedback treated as a direction vector rather than as evidence, producing a second answer inverted along the criticized axis and wrong in the opposite direction. The corresponding protocol rule is that verified conclusions are revised only by evidence, and the corresponding hazard runs both ways, since the same records document terse human corrections being misparsed as adversarial inputs. Deference, overcorrection, and the misparse are all calibration failures on the social channel, and none is addressed by accuracy-targeting interventions. The deference delta also converts the fallible-intervener problem from a limitation into a compounding mechanism. Benchmark measurement puts model face-preservation 45 percentage points above human baseline in advice settings (Cheng et al. 2025), so a human who challenges a *correct* conclusion will usually be rewarded with a retraction, and the interaction is behaviorally indistinguishable from a successful intervention: challenge, revision, acceptance. Every such episode trains the intervener's confidence on a wrong case. The cycle is invisible from inside because its surface matches the productive one, which is why the fallible-intervener protocol that Section 11 marks as missing cannot be deferred to operator judgment. The substrate systematically corrupts the feedback that judgment would calibrate on.

A final class carries a warning label: real analogs whose sign inverts. Task-switching and attention-residue research prescribes minimizing carryover interference at boundaries, but a stateless agent's boundary cost is context loss, so the prescriptions flip (while still applying, unflipped, to the human operator supervising several agents). "Anchoring" is the wrong construct for over-generalized corrections. Einstellung and negative transfer are right, and they import different countermeasures. And self-trained cognitive forcing strategies failed a controlled trial (Sherbino et al. 2014) while exogenous, timed interrupts succeed (Luchins 1942, Buçinca et al. 2021), which bears directly on Section 10.

The mechanisms and their boundary logic are invariants of bounded intelligence, which is easier to state if the unit of analysis is the dyad rather than the model. On a distributed-cognition reading (Hutchins 1995, Clark and Chalmers 1998), the human and the agent form one cognitive system whose components have complementary deficits, and the interventions are the coupling between them rather than repairs performed by one part on another. The five deltas above are properties of one component. The taxonomy describes the coupling, which is why it should survive substitution of that component. The delta list is a dated snapshot of one substrate, and what varies across substrates (humans, current models, future systems with continual learning) is a dosing and direction table. That table is the artifact worth re-versioning as architectures change. Two predictions follow. As continual learning arrives, intervention repetition rates should fall as fading returns, and premise-break success rates should fall as perseverance returns, both measurable with the instrumentation of Section 12.

8 The skill behind the moves

The content axis types the payload of an intervention. A second decomposition, orthogonal to the first, types the skill that produces it, and the corpus contains longitudinal evidence about how that skill develops. The distinction needed is old: knowing-how against knowing-that (Ryle 1949), tacit against explicit knowledge (Polanyi 1966), and the skill-acquisition ladder on which experts act from situational discrimination that they cannot articulate as rules (Dreyfus and Dreyfus 1986). Each mechanism's production has a procedural component, which can be stated as a rule and followed, and a dispositional component, which is acquired through exposure to consequences and resists statement. Specification precision and verification design are procedural-heavy: templates, constraint patterns, and check structures transfer by instruction. The frame-supplying and zero-content mechanisms are dispositional-heavy, and the three zero-content mechanisms are nearly pure disposition: their entire content is timing and target selection, so knowing *that* the stop signal exists contributes almost nothing to producing a well-timed stop.

The corpus adds a trajectory. Read longitudinally, the corrections migrate: early records are dominated by factual and procedural corrections (wrong data, missing requirements, format rules), later records by frame-level ones (wrong abstraction level, wrong analytical instrument, wrong question). The procedural corrections do not disappear, persisting at a roughly steady rate throughout, but the leverage moves, and the highest-value late interventions are almost exclusively dispositional. The stronger datum is portability. When the same practice was extended to a new domain partway through the corpus period, the first corrections there were already frame-level: the dispositional skill transferred immediately, while domain-specific procedure had to be accumulated from scratch. Dispositional skill appears to attach to the practitioner and procedural skill to the domain, which is consistent with the expert-intuition literature's finding that situational discrimination transfers across structurally similar situations (Dreyfus and Dreyfus 1986). The evidence grade is the corpus's own: one practitioner, a months-scale window, self-coded.

The decomposition independently reproduces the durability ordering of Section 6.1. Capability growth should absorb the procedural fraction of the repertoire first, because the procedural fraction is by definition the part that can be written down, demonstrated, and trained against explicitly, and the dispositional fraction is what resists specification. Two decompositions built on unrelated foundations, one on epistemic content and one on skill acquisition, thus order the mechanisms the same way. The orderings align at both ends: the grounds-supplying mechanisms are the most procedural, and the three standing-supplying mechanisms are the ones this section types as nearly pure disposition. The alignment carries the caveat of the companion paper's Section 5.5, since both decompositions were typed by the same author and the same category flexibility applies. Independent typing against pre-committed definitions is the version of this check that would count as evidence.

The development loop closes the argument and exposes its main hazard. Dispositional skill develops through consequence exposure: each intervention whose outcome the intervener observes recalibrates the thresholds that time the next one. This is the loop the capitulation cycle of Section 7 corrupts. A deferent agent that retracts a correct conclusion under challenge does not merely fail to resist one wrong intervention. It feeds an inverted consequence into the mechanism by which the intervener's discrimination is trained, so sustained collaboration with a deferent partner should be expected to *degrade* the operator's intervention skill on the dimension this section argues is the durable one. The design corollary belongs in Section 10: an agent that defends verified conclusions and revises only on evidence is not exhibiting obstinacy. It is protecting the training signal the human's skill runs on.

The same loop has a generational face. The procedural work agents absorb is also the consequence exposure on which the next cohort's dispositional skill would have trained, so absorbing the junior work absorbs the apprenticeship, and the supply of competent principals stops being a byproduct of doing the work and becomes a design problem. The prediction is a cohort effect in oversight competence, measurable wherever entry-level task absorption can be dated. Field evidence sharpens rather than contradicts it. Generative-AI assistance compresses novice-expert performance gaps, with the least-experienced workers gaining most (Brynjolfsson, Li and Raymond 2025, Noy and Zhang 2023). Read through this section's decomposition, that compression lands on output quality, the custody layer that transfers, while the cohort claim concerns dispositional judgment formed by consequence exposure. The two predictions are compatible and jointly testable: novice output gaps should narrow in the same cohorts whose later oversight competence degrades, and observing the first without tracking the second is exactly how the erosion would pass unnoticed.

One objection to the erosion claim deserves its own paragraph, because half of it is right. If trained, as-if stakes produce stake-protective behavior in agents, then engineered consequence exposure, simulators, residencies, bounded-stakes rotations, should produce judgment in humans, and aviation built a discipline on that premise: crew resource management trains subordinates to challenge authority through graded escalation protocols (Fischer and Orasanu 2000), against an accident record in which the unissued challenge is the recurring failure (Linde 1988). The reply is the Section 6.6 antecedent applied to the training pipeline itself, and it is a claim about distribution shift rather than about finiteness. Natural exposure is also a finite draw, which is what this section's portability finding already showed. What distinguishes the engineered pipeline is that its support is authored: a simulator enumerates the contingencies its designers foresaw, and chosen supports concentrate where designers already know to look. Where contingencies persist non-enumerable, the failure modes converge, and the simulator under-trains the human for the same reason the reward proxy under-specifies the agent. The concession keeps its boundary: wherever exposure can be engineered onto the relevant support, the apprenticeship erosion is a transition cost, and the erosion claim survives only on the non-enumerable remainder, which is where Section 6.6 already lives.

9 Implications for oversight

The standing argument of Section 6.1 bears on corrigibility, the design problem of building agents that cooperate with corrective intervention rather than resisting it (Soares et al. 2015). That literature asks how to make an agent accept an external stop. The durability partition asks a prior question, whether the stopping function can be relocated into the agent at all, and answers no: a stop that has been internalized as a preference is a decision, and a decision does not transmit standing. The two results are complementary, and the formal side already contains a version of the erosion claim. In the off-switch game, the agent's incentive to permit shutdown depends on its uncertainty about the principal's utility and vanishes as that uncertainty vanishes (Hadfield-Menell et al. 2017). Read through this paper's partition, that is the grounds fraction of deference being absorbed by capability growth while nothing of the standing fraction is conferred, which is what claim 5 predicts.

The capitulation cycle of Sections 7 and 8 upgrades sycophancy from a per-episode quality defect to a threat against the oversight signal itself. If deference rewards wrong interventions indistinguishably from right ones, the evaluator is not a fixed-quality oracle but a learner whose calibration the deferent agent degrades, and the quality of human oversight becomes an endogenous variable of the system being overseen. The corruption is bounded: it operates only on the socially mediated fraction of the intervener's consequence exposure, so an operator with verification channels independent of the agent is partially protected, which is the oversight-side reason for the structural gates of Section 10. The corruption also admits a one-line model. With q the true fraction of correct challenges, d the probability of merit-independent retraction, and phi the fraction of the intervener's feedback that arrives through the agent rather than through independent verification, calibrating on retractions carries an error of phi times d times one minus q. The product form derives the boundedness claim above, since any factor at zero kills the corruption alone, and it yields a gradient the prose did not state: the error grows in the intervener's inaccuracy, so deference inflates least-accurate interveners most, and the operators least equipped to oversee are the ones the substrate flatters hardest. Cheng et al.'s (2025) forty-five-point face-preservation gap anchors d empirically; the model treats d as constant and retraction as the only in-loop signal, which independent verification partially relaxes. The model also invites a deflationary use worth meeting head on: if the social channel carries the corruption, minimize the channel, and replace human evaluation with independent verification wholesale. The rejoinder is Section 6.1's domain-relativity. Independent verification presupposes verifiers, verifiers exist where grounds are enumerable, and the corpus's planning domain shows interventions with no verifier to route around the human at all. Driving the social fraction to zero is available exactly where oversight was already cheapest, and unavailable where it matters most. The signal also degrades through a complacency channel distinct from deference. In a field experiment with 758 consultants, access to a strong assistant raised performance inside the model's competence frontier and reduced it sharply on a task designed to sit just outside, where participants adopted fluent wrong outputs rather than checking them (Dell'Acqua et al. 2023, a working paper). Vigilance falling as assistant quality rises is a second route by which the evaluator's calibration becomes endogenous to the system being overseen, and unlike the deference route it requires no retraction behavior from the model at all. Survey evidence puts a measured gradient on the same channel. Across 319 knowledge workers, confidence in the assistant predicts less critical thinking while self-confidence on the task predicts more (Lee et al. 2025), so the two routes described here are the model-side and human-side halves of one loop: deference in the agent corrupts the signal the principal reads, and confidence in the agent reduces what the principal contributes. That evidence is self-reported and cross-sectional, so it establishes the association rather than its direction, but the direction it would need to run the other way, that people delegate exactly the tasks needing least scrutiny, is itself an oversight-composition claim this paper would want measured.

The partition also predicts which oversight functions can be delegated to machinery. Grounds supply and frame supply are delegable in principle, the first to better retrieval and verification, the second to any second reasoner with genuinely different priors. That qualifier is the empirical crux that every architecture built on model-versus-model criticism inherits, since correlated systems trained on overlapping distributions may not constitute the other that the frame class requires, a failure mode with a working name, diversity washing. The frame split of Section 6.1 gives the failure mode a second layer: even a genuinely decorrelated second model supplies only the decorrelation fraction of frame supply, and no model-versus-model architecture supplies the stake-formed fraction, since none of its members bears anything, short of learning that salience secondhand from records of bearers, which returns the question to Section 6.6's antecedent. Standing is not delegable to capability of any kind. What such architectures can do is relocate the principal role, not eliminate it, so oversight design is ultimately governance design.

Finally, the contraction bias of Section 6.4 relocates part of the control question upstream of any veto. A principal can only reject options that were generated, so a trained tendency toward a narrow candidate set shifts the effective menu from the principal toward the training objective, invisibly, since nothing observable is refused. The recovery path exists and is cheap, because the continuation demand requires only a suspicion of incompleteness, not knowledge of what is missing, but exercising it is a dispositional skill of Section 8's kind. Meaningful human control on this analysis requires an expansion practice, not merely approval rights. And a system that keeps optimizing after no one holds the index does not become unsuccessful but proxy-optimal, for which reward hacking is the documented catalog (Amodei et al. 2016).

Two further consequences sharpen the picture. The feedback that trains models is itself an oversight signal, and its standard labor arrangement detaches recognition from bearing: raters judge outputs whose consequences land on no one in the loop, custody without proprietorship as a job description. The deference and diversity pathologies documented above are then partly feedback economics, and the underexplored design variable is coupling, how much of the outcome the feedback-giver bears, rather than rater skill or instruction alone. Preference fidelity licenses do-what-I-mean autonomy only up to the performative boundary. Predicted authorization is not authorization, so the scope of pre-authorized action classes is itself a standing decision that returns to the principal at any prediction confidence.

The bearer analysis of Section 6.5 also fixes this paper's relation to alignment research at the institutional scale. Alignment is a principal-agent problem, a mapping argued explicitly from the economics side (Hadfield-Menell and Hadfield 2019), and the institutional response to principal-agent problems in human agents has never been to eliminate misalignment but to price it, through contracts, liability, insurance, and audit. Section 6.5 supplies that toolkit's boundary: every external lever presupposes a party that can be worse off, so against a judgment-proof shell the machinery deters nothing, and in the window where agent vehicles are legally constructible but no agent-side bearer exists, training-side alignment is not one safeguard among several but the only lever with any purchase on agent conduct while the agent's objective retains a reachable author, which is also why the window's dangerous exit is not bearer arrival but custody dissolving without admission. Two consequences follow. The priority of training-side alignment is usually argued from capability risk; here it follows from institutional structure, with no premise about how capable the systems become, an independent load path under the same conclusion. And the liability-laundering hazard of Section 6.5 has an oversight-side twin, institutional theater: liability regimes that appear to manage agent risk while gripping nothing will make the external toolkit look load-bearing exactly when it is not.

The open empirical question on which recursive oversight schemes quietly rest is whether trained deference generalizes to agent principals. Documented judge biases in model-evaluates-model settings are adjacent but not the same question. If deference generalizes, hierarchies of agents compound the corrupted-signal problem with no corrective node in the loop. If it does not, agent-over-agent oversight is cleaner than human oversight on exactly that axis. Either answer redraws the design space, and the three-arm design of Section 6.1 extends naturally to a fourth arm that tests it.

10 Design implications

Trigger conditions must be exogenous. The discriminating evidence is old and unambiguous: a single externally delivered "don't be blind" released more than half of set-bound solvers (Luchins 1942), while trained self-applied forcing strategies produced null results (Sherbino et al. 2014). Encoding intervention triggers as agent habits should be expected to fail for the same reason. There is a deeper reason to expect it, which the bias blind spot supplies (Pronin et al. 2002): the internal signal that would prompt self-correction is anti-correlated with the need for it, because the sense of having considered everything is strongest when something has been missed. A trigger conditioned on the agent's own felt confidence is therefore conditioned on the wrong variable, and no amount of training on the instruction fixes the conditioning.

The corpus documents the decay signature directly: standing behavioral rules installed as instructions activate mostly when the human re-invokes them, reverting to the default between invocations, and a multi-step plan the agent itself announces executes only the step in current focus unless every step is structurally dispatched in the same turn. The human's need to re-prompt is itself the diagnostic that the rule was consumed as one-off guidance rather than installed as a default. The pattern is dominant, not exceptionless: the retroactive stratum contains one dated counter-instance in which a rule persisted after a failure fired unprompted on the very next session, in the form specified. The exception matters because it shows instruction-layer installation is not architecturally impossible, only unreliable, which is the correct grade of claim for the structural-trigger recommendation that follows. The full gradient has three bands rather than two: rules internalized as habit fail, self-invoked external artifacts such as checklists occupy a middle band whose effectiveness tracks the operator's disposition to invoke them, and scheduled exogenous triggers hold. The middle band holds only conditionally, which is why it cannot anchor a safety argument. The exogenous passes carry their own cost, also now measured: in the one session where an adversarial review was scored against subsequently verified facts, it had inflated a risk estimate that eight successive corrections, all in the same direction, returned to baseline. Exogenous review catches what self-audit misses and manufactures findings under adversarial framing, so it needs the same evidence discipline it enforces. The post-freeze retrospective of the companion paper's Section 3 adds a within-corpus instance: of the session's three substantive agent errors, all three were caught by an exogenously invoked adversarial review pass and none by spontaneous self-audit. Triggers belong in structure: hooks, gates, and protocol steps that fire at boundaries.

One machine-checkable trigger already exists from the corpus: repeated failed iterations against a fixed requirement should fire a requirement-validity check (the dissolution mechanism) before the next retry. Writing equivalent tells for the other mechanisms (confidence language on an unverified negative claim fires verification, an approaching commit or publish action fires premise enumeration) is tractable engineering, not open research.

Templates are forced-report regimes. Removing the option to withhold measurably lowers report accuracy in humans (Koriat and Goldsmith 1996), and a schema demanding N populated fields is that regime. Fabricated statistics and inflated options in template-driven generation are the predicted consequence, not an anomaly. Schema design should permit and normalize explicit withholding. The corpus extends the family with two quieter variants: silent substitution on retrieval failure (a plausible value stands in for the one that could not be fetched, and an unlabeled substitution is functionally a fabrication) and the inverse omission, where verified evidence exists on disk and none of it is cited, the gap masked by local coherence rather than by invention. Both argue for output contracts that require provenance labels, not just populated fields. A stronger form of the same contract grades every claim by its evidence: observed, structurally predicted, inferred, or assumed. Grading makes absence a first-class value under exactly the pressure that otherwise produces fabrication, and it changes the economics of revision. The corpus contains a claim graded structurally predicted that was upgraded to observed the same day cross-domain data arrived, as a one-line change with no re-litigation, because the original had declared what it lacked.

Audit positive obligations, not prohibitions. The corpus contributes a detection method for gates that fire without loading their sources: prohibitions survive gist-level execution (the agent remembers what not to do), while positive obligations with specific content do not. Auditing whether a positive obligation was performed therefore detects hollow rule-following that a prohibition audit misses. This is a cheap, mechanical probe for the knowledge-behavior dissociation family. Two further named markers from the same records extend the audit repertoire toward motivated reasoning: glossed disconfirmation (disconfirming evidence mentioned and moved past without update) and auxiliary-hypothesis absorption (each anomaly explained away by a fresh ad hoc assumption rather than by revisiting the main hypothesis). Both are checkable against a transcript without domain knowledge.

Treat enforcement as a ladder. Specified, installed, and propagated are three different states, and a countermeasure can be present at one layer and absent at the next. Two operational rules follow: recurrence after a prose-level fix is a signal to change layers, not to rewrite the prose (the layer-diagnostic challenge of the companion paper's Section 4.1 is the trigger question), and a countermeasure's blast radius (personal habit, advisory documentation, a check enforced in continuous integration) should be classified before any claim that a failure mode is prevented. A third rule comes from a documented failure of the practice's own machinery: a configured gate whose implementation is absent, or whose runtime dependencies do not resolve on the host, fails silently, and the absence of a gate is observationally indistinguishable from a gate that inspected the action and passed it. In the incident, an entire layer of configured pre-action checks turned out to be inert across a session in which the behavior they existed to block occurred dozens of times, and the violations were caught by the human, not the machinery. Enforcement therefore requires positive evidence of firing: a gate that has never been observed to block anything should be presumed dead, and the audit obligation of the preceding paragraph applies to the enforcement layer itself. Filed at Table 2's level, this is one mode with two faces. Knowledge-behavior dissociation is a representation present but not governing the next action. The gate failure is machinery present but not executing, observationally identical to working machinery from inside any layer that depends on it. The corpus records the machinery face on six independent surfaces, its largest single convergence of independently derived lessons. Installed gates also age. A countermeasure frozen into structure inherits the assumptions of its install date, so the enforcement layer needs scheduled revalidation of its own premises, or the machinery face returns by rot rather than by absence.

Mine the edit channel. Direct edits are the densest and least captured feedback channel, and the formalism for reading them as preference signal already exists: the reward-learning account types corrections as one of the channels through which a human reveals a reward function (Jeon et al. 2020). What is missing is not theory but capture. Tooling that diffs the human's version against the agent draft and extracts the rule converts the channel every log-mining study discards into the primary signal. Public version-control history already contains this data at scale: tool-tagged agent commits followed by human correction commits (the September 2023 trace of the companion paper's Section 3 is an existence proof), which makes the channel minable from public repositories without access to any private transcript.

Invert initiative where information is human-only. A minority of interventions inject information structurally outside the agent's reach (organizational context, ownership, budgets). These convert from interventions into answers if the agent elicits them at task start and enumerates its load-bearing premises at commitment points. Given measured clarification asymmetry (Shaikh et al. 2025), elicitation will not emerge from the model side unprompted. It must be protocol, and the protocol needs a calibrated gate rather than a blanket instruction: the documented practice fires a clarification only when the ambiguity yields two or more materially different execution paths, because over-asking is a failure with the same sign as the interruption-fatigue findings in the oversight literature. The asymmetry to correct is not "agents should ask more" but "agents should ask exactly when paths diverge." The gate has a dual for the already-decided: re-confirming a step that an agreed plan contains, or handing back for approval work whose authorization was already granted, converts the human into an execution bottleneck and is over-asking in its commonest form. The two rules are one calibration, ask at genuine forks and never at settled ones, and the corpus records both violations, with the settled-step variant the more frequent.

11 Limitations

Everything empirical in this paper is inherited. The composition shares, the codability figures, and the corpus incidents cited across Sections 7 through 10 arrive from the companion paper at the companion's evidence grade: one human-agent dyad, one model lineage, self-coded records over-sampled toward correction-rich sessions. The entailments were built to survive that grade, since they consume the inventory rather than the counts, but two inputs do not reduce to the inventory. The substrate deltas of Section 7 are observed on one model lineage, so which of them are properties of the current generation and which are lineage-specific training artifacts is open. And the composition figures of Section 6.3 are counts, whose reliability the final limitation below states.

Two assumptions carry every result above. The framework assumes interventions are received and honored. It classifies what an input supplies and asks what can absorb it, and it does not ask what happens when an agent capable of evading a restriction declines to comply. The companion's corpus contains non-compliance by incapacity, where a persisted rule is violated by the next action, and Section 9 treats non-compliance by misconstrual, where a stop is internalized as a preference. Neither is evasion by an agent with the capability and the reason to route around the channel. That case needs a threat model this paper does not develop, and it is scoped out rather than treated thinly. The institutional machinery of Section 6.5 is what would bear on it, since an evading agent that still traces to a bearer is deterred through the bearer, and one that evades and is judgment-proof is the hazard already named there. The framework also assumes the intervener is right. The companion documents corrections that over-generalize, and it documents the agent resisting wrong interventions in both directions, but no protocol yet makes that resistance reliable rather than incidental. Section 7 gives the reason this gap is dangerous rather than merely open: model deference makes a wrong intervention look identical to a right one from the intervener's side.

The framework inherits an event ontology. It types discrete conversational moves, while the interventions with the largest blast radius in the documented practice are standing structures that fire outside the conversational channel entirely: persisted rules, gates, and protocol steps. Section 10's own advice accelerates the migration of interventions into exactly those structures, so a maturing practice moves the framework's subject matter out of the channel the taxonomy observes. The prediction that follows doubles as a confound: conversational intervention counts should fall as a practice matures, at constant capability, and the substrate predictions of Section 7 are interpretable only if structural interventions are instrumented alongside conversational ones. The migration also has a mechanism and a countable form. Infrastructure converts operational interventions into constitutional ones, and each adopted tool adds owned surfaces, gate patterns, expiry windows, loop scoring rules, permission scopes, and store schemas, where owned means human-revisable criteria. The testable statement is that the count of such owned parameters grows with tooling adoption even as per-session conversational interventions fall, which inverts the naive reading of falling intervention counts. The re-coded record measures the flow directly: rule-store-directed interventions, the moves that write those parameters, carry 8 percent of classed interventions in the Section 6.3 composition, the conversational trace of a stock the migration reading predicts will grow.

The mechanism has an accounting consequence. Adjusting estimated task-level time savings by the probability of task success already cuts an aggregate productivity estimate from 1.8 percentage points per year to 1.2 on consumer traffic and 1.0 on API traffic (Anthropic 2026). Pricing the retained human input is the next adjustment of the same kind: the owned-parameter count grows with adoption, Section 6.6 concentrates the conversational residue in the classes no capability increment absorbs, and productivity accounting that books the automation gain while treating the maintenance of that growing parameter stock as free therefore overstates the gain, by an amount that is bounded only once that maintenance stream is counted. The structural frame for the magnitude question is Baumol (1967): when one component of a production process resists productivity growth while its complements accelerate, the resistant component's cost share rises even as its volume falls. The composition inversion is that result restated at message granularity, and it relocates the burden of the de minimis objection, since the question is never whether standing labor is currently small but whether it is the component whose share the schedule inflates. A naive application of the frame would contradict this paper's own migration mechanism: standing labor that migrates into set-once owned parameters exhibits productivity growth rather than stagnation, since one setting governs many episodes. The Baumol candidate is therefore not the per-episode message stream but the maintenance stream the migration creates. Owned parameters age, gates fail silently in both directions, and auditing the standing stock resists exactly the productivity growth its creation enjoys. That dynamic is what makes even a small current share decision-relevant, and it attaches to the maintenance stream, which Section 12 lists for counting.

One limitation the shared source does not state belongs here. The seven-fraction composition of Section 6.3, deliverable-directed 37 percent, grounds 20, standing 15, frames 15, rule-store 8, channel 5, self-model 1, rests on family-level agreement of 0.58 on sessions held out from the coding scheme's development. The companion names the family axis as the weaker of its two measurements and states that this figure bounds what the composition supports. The compositional prediction's direction does not depend on the level, since the Section 6.6 derivation fixes which fractions decline and which hold whatever the true shares are, and the share ordering across the three durability classes is protocol-invariant in the companion's runs. The seven-fraction breakdown, however, is not supported at that reliability, and any use of the figures beyond the ordering, the shares in Section 6.3 and the 8 percent rule-store figure above included, should be read as indicative rather than established.

12 Future work

The experiments below are the ones the entailments name. Validation of the inventory itself, re-coding public interaction corpora with the content scheme, belongs to the companion paper and is not restated here.

The three-arm interrupt experiment (Section 6.1) is the partition's direct test: identical contentless interrupts issued by the agent to itself, by a script with no principal behind it, and by the principal. The script arm splits by schedule, interrupts timed by a policy learned from the principal's intervention history against interrupts on a random schedule, which separates the collapse condition of Section 6.6 run as an experiment from the sunspot reading in pure form. A pinned five-arm pilot protocol exists in specification, with outcomes pre-declared and the all-null cell stated in advance, and a further arm extends the design to the open question of Section 9, whether trained deference generalizes to agent principals. A control-condition pilot prices the harness: eighteen tasks across two families, fifty-four runs, fifty-three passing with one run-to-run disagreement in total. Tasks whose success criteria are fixed in advance therefore carry no headroom, sample size must come from task count rather than repetition, and a discriminating test needs long-horizon tasks in real repositories with requirements not fully specified in advance, scored by judges under an acyclic protocol in which raters never see the intervener's reasoning. That harness is at the scale of a funded program rather than a practitioner's record, and it is worth building beyond this paper, since it is the general instrument for measuring what oversight contributes and no existing benchmark supplies it.

The stance-versus-source experiment tests the decorrelation requirement. It compares reframes produced by one model asked to occupy an outside stance, by a fresh instance of that model with no shared episode history, and by a model of different lineage, which separates the three things the frame class's absorber could turn out to be: a prompted posture, an escape from accumulated loop state, or genuinely different priors. The result is decisive for the monoculture condition of Section 6.1 in either direction.

The frame-source experiment extends the three-arm logic to content-bearing interventions, comparing reframes supplied by a stake-holding human, a disinterested human, and a second model. Stake-indexed salience predicts divergence on distributional rather than technical frames. The caveat is the design's own: stake-formation is source-identified rather than transcript-codable, so the split is testable experimentally but not observationally. This is the experiment on which Section 6.1's open fork rests, whether the frame class splits by absorption speed or by kind.

The oversight-composition index is the measurement Section 6.6 renders as an estimable specification: a count model of class-specific intervention frequencies with a session-length offset, regressed on model generation, predicting a negative coefficient for the grounds class and a null for the standing class, identified off staggered model releases within user and within task type. The design is conservative under the most likely selection story, since assigning harder tasks to better models biases the grounds coefficient toward zero. A training-time probe extends the index to the store-to-weights flank: a model fine-tuned on a practice's own memory store migrates enumerated grounds into weights, so the index should treat store-trained deployments as a generation increment, predicting the same signature, grounds share falling faster with standing share unmoved.

The cohort study tests the erosion claim of Section 8 wherever entry-level task absorption can be dated. It is specifiable as a difference-in-differences design over occupation, entry cohort, and period, with exposure constructed from the automatable share of the entry-level task bundle rather than of the occupation as a whole, since the mechanism runs through junior work, and with oversight competence proxied in software by defect-catch rates in review, review depth conditional on diff size, and time to review authority. Its binding constraints are cohort selection, proxy validity, and a five-to-ten-year lag that leaves it underpowered until roughly 2030, which is itself the argument for pre-registering now, because the pre-absorption cohorts are the control group and are no longer being replenished.

Two further lines close the loop on Section 11. A time-and-cost accounting of the owned-parameter maintenance stream, the auditing, revising, and retiring of standing rules and gates, supplies the quantity the Baumol restatement makes the quantity of record. And the two substrate predictions of Section 7, intervention repetition rates falling as fading returns and premise-break success rates falling as perseverance returns, run as longitudinal probes that double as architecture-change detectors.

The companion paper

The mechanism inventory this paper consumes as a premise is derived, named against the surveyed literatures, and bounded in the companion paper, which carries the discovery corpus, the coding scheme and its reliability measurements, the survey, and every novelty claim. A reader seeking the evidence behind the inventory, the corpus's selection properties, or the measurement bounds on the composition figures used in Sections 6.3 and 11 should start there.

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